

GEEM 433

ENGINEERING ETHICS AND MORALE
PHILOSOPHY



Marks Distribution at a glance

- Total Available Points:
- Class Tests:
- Midterm Exam:
- Final Exam:
- Case Studies:
- Group Presentation:



What will be covered in this course

Course Content

- Engineering Ethics:** Introduction to Ethics; Theories of Ethics; Principles of Engineering Ethics; Ethical expectation: Employers and employees, Inter-professional relationship,
- Standards and codes:** Fundamental Canons, NSPE (National Society of Professional Engineers) codes, IEEE codes of conduct, ACM codes; Institutionalization of ethical conduct. Ethical Dilemmas, Choices (Whistleblowing),
- Computer Ethics:** Computer Crime and Cyber Security, Privacy and Confidentiality issues in CSE, Legal Framework in CSE-Copyright laws, ICT Act, Right To Information (RTI), Patents, and Royalty etc. Ethical Challenges for CSE Engineers with the Advancement of Technology;
- Case studies** related to ethical issues in ICT and other Engineering disciplines.
- Introduction to **Philosophy of Engineering**, metaphysics, epistemology, axiology, and logic.

What will be covered in this course

- This course motivates engineers to perform under a standard of **professional behavior** that requires compliance to the **highest principles of ethical conduct** and **managing resources and decisions** effectively.
 - Part of professional ethics is the understanding of the ethics of other professions: how they **interact** and what can be **expected** from them as correct ethical behavior. It **improves** the profession, helps **set better rules** for the future, and **shapes** the way people think and act ethically.

Scenario:

- An engineering firm is hired to design a smart **healthcare monitoring system** in partnership with a hospital. The engineers must work closely with medical professionals and data privacy officers.
- Professional Behavior & Ethical Conduct?
- Outcome?

What will be covered in this course

Scenario:

- An engineering firm is hired to design a smart healthcare monitoring system in partnership with a hospital. The engineers must work closely with medical professionals and data privacy officers.
- Professional Behavior & Ethical Conduct:
 - The engineers ensure patient data is encrypted
 - They respect the doctors' ethical obligation
 - Avoid unethical vendors
- Outcome:

By understanding and respecting the ethics of the **medical profession**, the engineers contribute to a solution that is **technically sound, ethically responsible, and socially beneficial**.



What will be covered in this course

- **Objectives**

1. To develop a firm ethical base.
2. Understanding Legal Issues & Professional Development
3. Assessing Professional Conduct Codes
4. To identify and analyze practical legal problems commonly encountered in the computing industry.

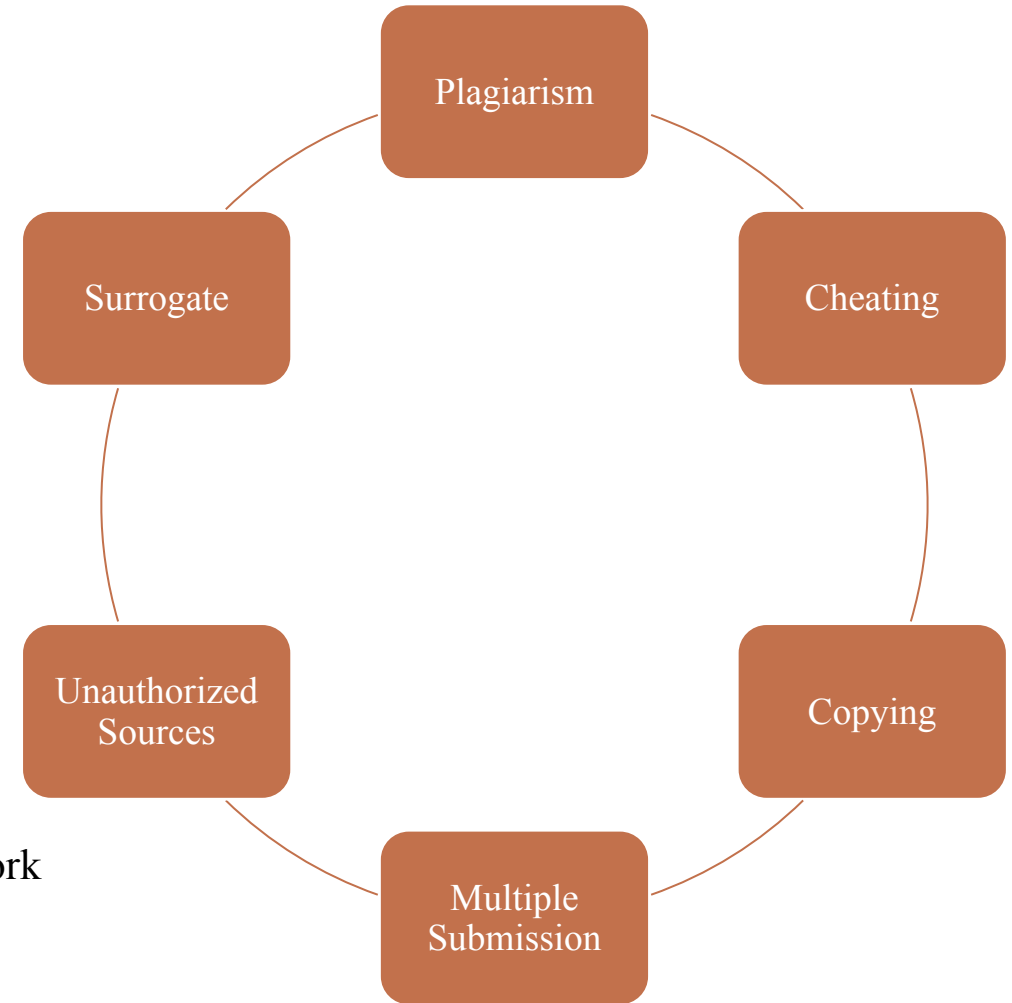
What Does Ethics Mean?

- Ethics refers to the principles and values guiding individuals or groups in determining right or wrong, good or bad, just or unjust, virtue and vice. It helps people decide how to act in various situations, emphasizing fairness, respect, and responsibility.



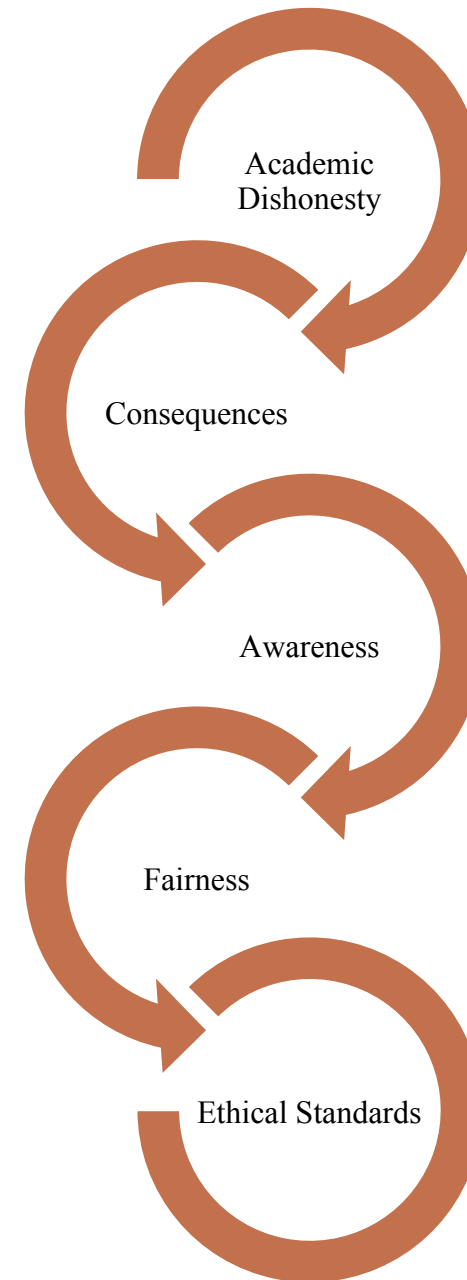
Student Code of Ethics

- ❑ **Plagiarism**
Using someone else's words, code, or ideas **without credit**.
- ❑ **Cheating**
Using **unauthorized help or resources** during exams or assignments.
- ❑ **Copying**
Submitting work that was **copied from a classmate** or letting someone else copy yours.
- ❑ **Multiple Submission**
Submitting the **same work for more than one course or assignment** without permission.
- ❑ **Unauthorized Sources**
Using **external services or people** to complete your academic work without permission.
- ❑ **Surrogate**
Having **someone else complete work or attend exams** on your behalf.



Student Code of Ethics

- What should we do to maintain the highest Ethical Standards?
- **Avoid Academic Dishonesty**
Don't cheat, plagiarize, or use surrogates. Always submit your own original work.
- **Understand the Consequences**
Violations can lead to failing grades, disciplinary action, or expulsion.
- **Raise Awareness**
Know what counts as unethical behavior (e.g., copying code, unauthorized collaboration).
- **Practice Fairness**
Treat others' work and efforts with respect. Don't gain an unfair advantage over classmates.
- **Uphold Ethical Standards**
Follow your institution's code of conduct. Make ethical decisions even when no one is watching.



Ethics

- **Why do we need it?**
 - Ensure harmony of the group while providing mutual benefits to the individual
 - Basis for how we should treat others
 - It overrides self-interests

Moral Reasoning

- Moral reasoning is the process of **thinking** carefully about what is right or wrong, fair or unfair, and good or bad in a situation — especially when it affects other people.
- It's a way of **making decisions based on values** like:
 - ✓ **Justice** (Is it fair?)
 - ✓ **Equality** (Is everyone treated the same?)
 - ✓ **Freedom** (Are people free to choose?)
 - ✓ **Health and Safety** (Will this harm or help someone?)

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Moral Reasoning

Why It's Important?

- Moral reasoning helps us:
 - ✓ Make **ethical choices**
 - ✓ Stand up for **what's right**
 - ✓ Understand **consequences**
 - ✓ Become **responsible citizens or professionals**

Social Responsibility

- **Social Responsibility** requires taking into consideration the **needs of society**.
- **Engineers** must consider designing, building, and marketing products that benefit **society**.
- **One main connection** between ethics and engineering comes from how **engineering products** and **processes** affect society and their impacts.
- Ethics is about doing what's **right**, and engineering affects society—so ethical engineers must care about **people, safety, fairness, and the environment**.

Social Responsibility

- **Thinking Beyond the Project**

Engineers must ask:

- "Will this product be safe?"
- "Is it environmentally friendly?"
- "Does it help people or harm them?"

- **Examples of Social Responsibility in Engineering:**

- **Designing energy-efficient buildings** to reduce environmental impact.
- **Creating affordable medical devices** so more people can get treatment.
- **Developing apps** that protect user privacy and data.

Ethics in Engineering

Ethics is part of engineering for two main reasons.

- a) Engineers need to be socially responsible when building products and processes for society.**
- b) Social responsibility requires professional responsibility through obligations and duties.**

Typical Ethical Issues Encountered

- Safety
- Acceptable risk
- Compliance
- Confidentiality
- Environmental health
- Data integrity
- Conflict of interest
- Honesty/Dishonesty
- Societal impact
- Fairness
- Accounting for uncertainty, etc.

Typical Ethical Issues Encountered

- **Safety**

Ensuring systems or products do not harm users.

→ *Example:* An engineer refuses to approve a bridge design that doesn't meet load requirements.

- **Acceptable Risk**

Balancing innovation with the risk it brings.

→ *Example:* Launching a self-driving car only after extensive safety testing.

- **Compliance**

Following laws, regulations, and standards.

→ *Example:* A factory follows environmental laws to limit emissions.

- **Confidentiality**

Protecting sensitive or private information.

→ *Example:* A software developer keeps user data secure and does not share it with advertisers.

Typical Ethical Issues Encountered

- **Environmental Health**

Considering the impact of engineering activities on the environment.

→ *Example:* Using eco-friendly materials in construction projects.

- **Data Integrity**

Ensuring data is accurate, secure, and not manipulated.

→ *Example:* A data analyst refuses to falsify research results under pressure.

- **Conflict of Interest**

Avoiding situations where personal gain affects professional judgment.

→ *Example:* A government engineer steps aside from approving a project submitted by their brother's company to avoid a conflict of interest.

- **Honesty/Dishonesty**

Being truthful in reporting, design, testing, and communication.

→ *Example:* Reporting flaws in a software system instead of hiding them.

Typical Ethical Issues Encountered

- **Societal Impact**

Thinking about how a project affects society.

→ *Example:* Developing affordable water purification systems to improve public health in rural communities.

- **Fairness**

Ensuring equal treatment and access for all.

→ *Example:* Developing AI algorithms that are fair to all demographic groups.

- **Accounting for Uncertainty**

Being transparent about unknowns, risks, or limitations.

→ *Example:* Clearly communicating the limits of a predictive model to users.

Code of Ethics – Some thoughts!

- **It is not a legally binding document.**
- **It is not something that engineers need to memorize.**
- **It is something engineers should understand and live by in their works**
- **However, in the beginning knowing the code is a guide to understanding how to apply it.**



GEEM 433

Engineers: Professionals for the Human Good

**ENGINEERING ETHICS AND MORALE
PHILOSOPHY**

What is the Public Good?

- “Engineers should hold paramount the **safety, health, and welfare of the public**, as the NSPE code states. Probably, the most **fundamental** term here and certainly the most **ambiguous** and **controversial** is **welfare**.”
 - The term **welfare** appears to have several equivalents in engineering codes, such as **well-being and quality of life**.
 - The Preamble **(introduction)** to the NSPE code says that “**engineering has a direct and vital impact on the quality of life for all people**. The code of the Association for Computing Machinery obligates **(compulsory)** its members to “**contribute to society and human well-being**.”

What is the Public Good? Cont'd

Code of the American Society of Civil Engineers affirms that engineers should utilize **“their knowledge and skill for the enhancement of human welfare and the environment”**.

Finally, part of the introductory statement of the code of the Institute of Electrical and Electronics Engineers states that its members recognize **“the importance of our technologies in affecting the quality of life throughout the world”**.

Assuming the equivalence of these terms, we shall take well-being as our term of choice and say that **promoting the well-being of the public is the primary responsibility of the engineering profession**

So, what is Well-being?

No doubt, engineers have always assumed that their work contributes to the human good or what we have now called human well-being. But, until recently, little explicit (clear) consideration has been given to this goal. It is even conceivable (understandable) that engineers may one day be asked in some formal way to determine the well-being impact of their work, just as they now are often asked to determine the environmental impact.

The mandate (command) to engineers to promote human well-being or quality of life in their professional work is clear, but more guidance about the nature of well-being is needed.

Well-being - Psychologist Martin Seligman

Five elements of well-being include:

1. Positive emotion,
2. Enjoyment of activities in which one can be absorbed,
3. Connection to something larger than oneself,
4. Accomplishment in projects or work, and
5. Positive relationships.

These elements are closely related to the concept of happiness.

These ideas, however, may be somewhat difficult to relate to engineering. One possible way around this issue which may sometimes be useful is to take advantage of the widely discussed **Capabilities Approach (CA)**.

Two important developers in economics Amartya Sen and philosopher Martha Nussbaum. **According to Sen and Nussbaum,**

- **We do not have to determine what well-being is, but rather step back a little and ask what conditions are necessary for the realization of some of the most commonly recognized elements of well-being, regardless of how individuals or even experts may define it.**

extra

- **Positive emotion** – Experiencing feelings like joy, gratitude, and optimism.
- **Enjoyment of activities** – Engaging in activities that allow deep focus and absorption.
- **Connection to something larger** – Feeling a sense of purpose or belonging to a greater cause.
- **Accomplishment** – Achieving goals or succeeding in work and personal projects.
- **Positive relationships** – Building meaningful and supportive relationships with others.

Well-being & Engineering

Astronaut Neil explained - Even though each of us may have our own concept of what comprises quality of life, we can probably agree that **certain living conditions are essential to a preferred quality in our own lives.**

If we look at the capabilities suggested by CA(Capabilities approach) writers that are most closely related to engineering, we get a clue **what some of these living conditions might be:**

- Having food, shelter, and water,
- Having satisfying human relationships (communication, the Internet),
- Having free movement and expression (highways, air travel, the Internet, telephone, etc.), and
- Having a satisfactory relationship to the natural world (environmental preservation).

Promoting Well-being – Prohibited Actions

Many rules in ordinary or nonprofessional ethics identify what we should not do. For example, **ethical guidelines prohibit actions like dishonesty, stealing, and murder.** Prohibitions are a prominent part of professional ethics, including engineering ethics. Approximately 80 percent NSPE code consists of rules that clearly or indirectly tell us what we should not do.

Examples of Prohibited Actions from the NSPE Code(next page)

Examples of Prohibited Actions from the NSPE Code

- **Do not reveal privileged information (II,1,c)**
- **Do not associate with dishonest professionals (II,1,d)**
- **Do not aid the unlawful practice of engineering (II,1,e)**
- **Do not accept compensation from two parties on the same project (II,4,b)**
- **Do not participate in governmental decisions related to your own work (II,4,d)**
- **Do not solicit work from a governmental body on which a member of your firm has a position (II,4,e)**
- **Do not falsify your qualifications (II,5,a)**
- **Do not give bribes (II,5,b)**
- **Do not be influenced by conflicting interests (III,5)**
- **Do not unjustly injure the reputation of another engineer (III,7)**

Promoting Well-being – Prohibited Actions

Many parts of the **NSPE code** don't seem negative at first but actually **focus on setting limits and rules**.

Section II.1.b states that **engineers shall approve only those engineering documents that conform with applicable standards**. In other words, engineers shall not approve engineering documents that are not in conformity with applicable standards.

However, this doesn't imply that engineers must approve every document that does meet the standards. There are likely **additional criteria** that must also be met for an engineering document to be approved.

Promoting Well-being – Prohibited Actions

There are several good reasons for the prohibitive(strict) tone of the NSPE code and many other engineering codes.

- First, it makes sense that professionals, like engineers, should **prioritize not harming others**. It's important to avoid causing harm before trying to do good.
- Second, these codes are mostly written as **clear rules that can be enforced through penalties by professional organizations or even the law**. It's easier to enforce rules that specify what not to do rather than rules that ask for more open-ended and positive actions.
- For example, a rule that tells engineers to avoid conflicts of interest is much easier to enforce than a broader requirement like ensuring the safety, health, and welfare of the public.

Extra -2nd point

- **Clear Rules are Easier to Enforce:** If a rule clearly states that engineers should not have conflicts of interest, it's straightforward to check if an engineer has followed this rule. For example, if an engineer is found to have accepted money from a company while also working on a project for that company, it's clear that they broke the rule.
- **More Open-Ended Requirements are Harder to Enforce:** On the other hand, if a rule says engineers should “[prioritize public safety](#),” it's much harder to measure if they are actually doing this. There are many ways to interpret what “prioritizing public safety” means, making it tricky to decide if an engineer has followed the rule or not.

Prohibited Actions as Prerequisite for well-being

Protection from harmful actions is an **essential prerequisite for well-being**, no matter how it is defined. Taking just three examples-

- (1) If engineers are dishonest, they can mislead clients.**
- (2) If engineers have their professional judgment corrupted by conflicts of interest, they may make biased decisions.**
- (3) If engineers do professionally incompetent work, it can lead to failures or accidents.**

In all these cases, clients or employers suffer because they don't receive honest, fair, and competent advice. This reduces their ability to use engineering services to achieve their goals.

Overall, when engineers avoid these harmful actions, it helps improve the well-being of the public.

Promoting Well-being – Preventing Harm to the Public

Engineers are obligated not only to abide by code prohibitions to avoid causing harm but also to actively **prevent harm caused by technology or by other engineers in some situations.**

Prevention of harm usually involves:

- (1) identifying and disclosing potential harms and
- (2) attempting to prevent them.

These actions, while mostly about stopping harm, also have a positive position. They often **involve courage and require effort to stand up and prevent harm** to the public. In any case, they both protect and promote well-being.

Promoting Well-being – Preventing Harm to the Public

Different engineering codes explain the duty to prevent harm in different ways. Some are more specific. For example, **Canon 1** of the **IEEE Code of Ethics** clearly says:

- Engineers must take **responsibility** for decisions that affect public **safety, health, and welfare**.
- They must also **quickly disclose** (inform) if they find any **risks** that could harm people or the environment.

This means engineers cannot ignore or hide dangers. They have a duty to act honestly and responsibly to prevent harm.

Promoting Well-being – Preventing Harm to the Public (Case 82-5)

The NSPE's Board of Ethical Review appeared to recognize the category of preventive action in its decision on case 82-5, which was submitted by one of its members. In this case, an engineer was terminated because he repeatedly protested his employer's actions, believing the employer was wasting taxpayer money on a defence contract. The Board cited section II.1.a. of the NSPE code operative at that time, which reads:

Engineers shall at all times recognize that their primary obligation is to protect the safety, health, property and welfare of the public. If their professional judgment is overruled, under circumstances where the safety, health, property, or welfare of the public are endangered, they shall notify their employer or client and such other authority as may be appropriate.

The Board also cited section III.2.b of the code at the time, which stated that Engineers shall not complete, sign, or approve plans or specifications that are unsafe for public health and welfare, or that do not meet accepted engineering standards. If the client or employer insists on such unprofessional conduct, they shall notify the proper authorities and withdraw from further service on the project.

Extra

HERE'S AN EXAMPLE TO ILLUSTRATE THIS PROVISION:

AN ENGINEER IS WORKING ON THE DESIGN OF A BRIDGE, AND DURING THE PROJECT, THEY DISCOVER THAT A KEY SAFETY FEATURE IS BEING IGNORED TO CUT COSTS. THE ENGINEER RECOMMENDS INCLUDING THE SAFETY FEATURE, BUT THEIR SUGGESTION IS OVERRULED BY THEIR EMPLOYER, WHO WANTS TO PROCEED WITHOUT IT TO SAVE MONEY.

ACCORDING TO THE CODE, IF THE ENGINEER BELIEVES THAT THIS DECISION ENDANGERS LIVES OR PROPERTY, THEY ARE OBLIGATED TO NOTIFY THEIR EMPLOYER. IF THE EMPLOYER IGNORES THE CONCERN, THE ENGINEER SHOULD INFORM A HIGHER AUTHORITY, SUCH AS A GOVERNMENT SAFETY AGENCY OR A PROFESSIONAL BODY, TO ENSURE THE PUBLIC'S SAFETY IS NOT COMPROMISED.

THIS RULE ENSURES THAT ENGINEERS HAVE A DUTY TO ACT IF THEIR PROFESSIONAL JUDGMENT IS IGNORED IN A WAY THAT COULD CAUSE HARM.

Promoting Well-being – Preventing Harm to the Public

The NSPE Code of Ethics has two important rules:

- Engineers must avoid harming the public.
- They should also actively try to prevent harm, when possible.

In **Case 82-5**, an engineer protested their employer's actions, believing they were wasting public money. The **Board of Ethical Review** said:

- The engineer had the **moral right** to protest, based on their personal values.
- But after the engineer informed the proper authorities, they were **not required** to keep protesting or fighting the employer.

This means, engineers should speak up when they see something wrong, but they are **not expected to keep fighting forever**, especially after they have done their duty by reporting the issue.

Promoting Well-being – Preventing Harm to the Public

One of the practical reasons behind this idea is that **continuing to protest** could lead to personal risks for the engineer, like:

- Losing their job (getting fired)
- Damaging their career
- Facing workplace pressure or harassment

The **NSPE Code of Ethics** recognizes this reality. It says the engineer's **duty** is to raise concerns and inform the proper authorities if public safety, health, or welfare is at risk. After that, if they continue to protest, it is based on their **personal choice or moral conscience**, not a strict professional obligation.

Promoting Well-being – Aspirational Ethics

Although engineering codes of ethics place great emphasis on the importance of refraining from certain kinds of behavior (prohibited actions) and engaging in behavior that prevents harm, such provisions do not adequately capture the more positive aspects of engineering.

We call this more positive component of engineering ethics **'aspirational ethics'**. Aspirational ethics are about doing more than the minimum required. They encourage engineers to actively improve human well-being, even when it's not specifically required by the rules. Some of these actions are necessary because the codes say so, but others go beyond what's expected. In the next chapter, these extra actions will be called **'supererogatory'**, meaning they are good to do but not required.

Aspirational ethics ask professionals to exceed the minimum standards of practice and behave in ways that cannot be enforced or covered by codes of conduct. This higher level of ethical behavior is related to moral values and virtues, and it entails more reflection and discussion than mandates(commands) and prohibitions.



GEEM 433

Engineers: Professionals for the Human Good

**ENGINEERING ETHICS AND MORALE
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Some Thoughts!

- A person's profession is a part of personal identity.
- Engineering is a profession.
- Engineering codes and guidelines from leaders in the field require engineers to work for the public good. This means improving people's well-being, welfare, and quality of life through their work.
 - *Promoting the well-being of the public*, not engaging in professionally prohibited actions, preventing harm to the public, and actively promoting the public's well-being.
 - *In designing for well-being*, engineers must keep in mind the social context of engineering and technology and the need for a critical attitude toward technology.

DRIVERLESS CARS!

It is easy to understand the advantages that driverless cars offer: reducing traffic collisions, providing better transportation for the elderly and disabled, lowering fuel consumption, and increasing traffic flow.

On the other hand, they raise many social, legal, and ethical questions. Perhaps the most obvious question is who should have responsibility for accidents.

The first fatal accident of a driverless car occurred in Williston, Florida, on May 7, 2016. A person in a Tesla driverless car was killed when a truck made a left turn in front of the car. The car went under the truck's trailer without braking, likely because neither the autopilot nor the driver saw the white side of the trailer against the bright sky.

DRIVERLESS CARS!

- Where should **moral responsibility** and **legal liability** lie in this case?

-> In this case, moral responsibility and legal liability are complex. Investigation revealed that the driver did not operate the Tesla according to the instructions and Tesla also did not have a system to check if the driver was ready to take over the control of the car when needed. There's also the question of how realistic it is to expect a driver to stay fully alert when an autopilot system is installed.

- Liability and responsibility are **not the only questions** raised by driverless cars.
 - How safe are they?
 - What kinds of information should be given to drivers before they purchase or use these vehicles?
 - How should the potential problems of hacking and terrorism be handled? (To handle the risks of hacking and terrorism with driverless cars, strong security measures are needed.)
 - What about the potential loss of driving-related jobs? Should there be retraining for other jobs?

Questions of Responsibility and Moral Issues

- Questions about **responsibility** aren't just related to driverless cars; they arise in engineering disasters like the **Challenger and Columbia space shuttle accidents**.
- There are important moral issues regarding how **technology** affects the **environment** and how **engineers** manage **risks**. These concerns matter to engineers not just because they design the technologies but also because being an engineer involves **professionalism**, which has a strong moral aspect.
- Engineers must think carefully about their **responsibilities** and the **impact** of their work on society and the environment.

The two components of professionalism are:

- (1) **expertise in a certain area (accounting, law, medicine, engineering, etc.) and**
- (2) **adherence to moral guidelines, usually laid out in a formal code of ethics.**

Failure in either of these areas means one is deficient as a professional.

Three Stages in the Development of Professional Identity

- **Independent Operator.** Professionalism is meeting fixed and clearly defined guidelines and expectations that are external to one's character.
- **Team-Oriented Idealist.** Rather than identifying professionalism with fixed rules and behaviors, professionalism is about meeting the expectations of other professionals, especially those who are admired.
- **Self-Defining Professional.** At this level, a person's values are fully mixed with their profession's values. Professionalism becomes part of who they are. This stage is often not fully achieved until mid-life.

What is a Profession?

- The use of profess and related terms in the Middle Ages was associated with the public profession of a way of life that carried strict **moral rules** (**religious** and ethical duties).
- By the late seventeenth century, the term "profession" moved away from its religious meaning and became more **secular**. It started to describe people having right **qualifications** and **skills** to offer valuable services to others, such as doctors, lawyers, or teachers.
- **Three approaches to professionalism are especially important in understanding the concept and can be useful in understanding professional identity:**
 - **Sociological Account**
 - **Social Contract Account**
 - **Morally Permissible Account - offered by philosopher Michael Davis.**

What is a Professionalism?

Sociological Account - Society gives professionals (like doctors, engineers, lawyers) special status because they serve the public with expert knowledge..

Social Contract Account - Professionals have an unspoken agreement with society: "We give you **trust**, **money**, and **respect** — in return, you behave responsibly and work for the public good."

- On the one hand, **professionals agree** to attain a high degree of professional expertise, offer good service to the public, and follow ethical rules in their work.
- On the other hand, the **public agrees** to let professionals enjoy above-average wages, have social respect and prestige, and enjoy a considerable degree of freedom to regulate themselves.
- The idea of such an agreement creates a strong sense of responsibility on a professionals or a developing professional

Morally Permissible - A profession is a group of individuals working in the same field who voluntarily choose to serve the public by pursuing a morally desirable goal, such as helping others or promoting the public good. Davis' definition emphasizes that a profession consists of more than one person and includes a public element, serving as a way for people to earn a living and occupy their working hours. People enter the profession willingly, and it focuses on achieving a morally beneficial goal, such as curing the sick or promoting the well-being of society.

Characteristics of a Profession

- **Extensive Training:** Requires a significant period of intellectual training, usually obtained at a college or university.
- **Specialized Knowledge:** Professionals possess knowledge and skills essential for the well-being of society.
- **Monopoly on Services:** Only qualified individuals can offer certain professional services, and they control how others can enter the field and get trained.
- **Workplace Autonomy:** Professionals take decisions based on their judgment, not just based on orders from others.
- **Ethical Standard:** A set of rules about right and wrong, usually written in code of ethics, to ensure public well-being.

Engineering as a Profession

Engineering is clearly a profession by all three accounts.

- First consider the Sociological Account.
- **Extensive Training:** Becoming an engineer requires high level of training in the college or university.
- **Specialized Knowledge:** Possessing knowledge and skills is vital for engineers for the well-being of society.
- **Monopoly on Services:** The engineering profession does not have complete control over the practice of engineering, because, in some countries, such as the United States, one does not have to be a registered professional engineer (PE) in order to practice engineering.
- **Workplace Autonomy:** Engineers often have more autonomy in the workplace than many non-professionals.
- **Ethical Standards:** Engineers have ethical codes that ensure public safety, sustainability, and ethical practices.

Engineering as a Profession

Engineering is clearly a profession by all three accounts.

Social Contract Account

- Engineers are highly skilled professionals who provide competent services. While engineers have ethical codes to follow, losing their Professional Engineer (PE) registration for unethical behavior doesn't necessarily stop them from working in their field. Unlike other professions where losing a license means you can't practice, many engineers aren't required to be licensed to continue their work.
- However, engineers still enjoy high wages and significant social status as part of the benefits of their profession.